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Mamdani System

Observations and Explanations

Serial No.	MIS Parts	Observations	Explanations
1.	Membership Functions (MFs)	Trapezoidal MFs are defined for position error (p_{err}), derivative error (d_{err}), and force (f). The ranges of the universes ($[-20,20]$ for p_{err}, $[-40,40]$ for d_{err}, $[-50,50]$ for f) are symmetric and overlap significantly. Overlapping MFs allow an input to belong to multiple fuzzy sets at once.	Overlap promotes smoothness: Inputs near the boundary of two sets activate both rules, ensuring gradual changes in force instead of abrupt jumps. Large overlaps can cause spikes: When many sets fire at once (e.g., at large initial angles), their aggregated outputs may strongly skew the centroid, producing temporary spikes in control force. Trapmf shapes produce wide plateaus, making the system more tolerant to noise but slightly slower in response because the centroid shifts less sharply.
2	Rule Base	The 5×5 grid of rules maps combinations of p_{err} and d_{err} to force outputs. The rule matrix is symmetric: large negative error with large negative derivative produces a strong positive force (f_{pb}) to push the pendulum back upright.	Rule density determines responsiveness: More rules and finer granularity improve accuracy but increase computational load. Diagonal dominance (positive position with positive derivative $\Rightarrow$ strong negative force) ensures stability by opposing motion. If rule strength is too aggressive (e.g., many “PB” outputs for moderate errors), the system can overshoot because too much force is applied.
3	Inference and Aggregation	AndMethod = 'min' and OrMethod = 'max' are used. AggregationMethod = 'max' combines the outputs of all fired rules by taking the maximum membership grade across fuzzy sets.	Min–max inference provides robustness to noise but can lead to sharper transitions when multiple rules fire strongly. During large disturbances, overlapping high-output rules create wide, tall aggregated fuzzy areas, pushing the centroid of the aggregated output set toward the extreme ($\rightarrow$ force spikes).
4	Defuzzification	The controller uses centroid defuzzification. The centroid is sensitive to the shape and spread of the aggregated fuzzy output.	Centroid provides a smooth continuous output (good for control of physical systems). However, when large overlapping areas appear (e.g.,

			high positive p_err and d_err), the centroid may shift suddenly toward high positive force $\rightarrow$ overshoot or oscillation.
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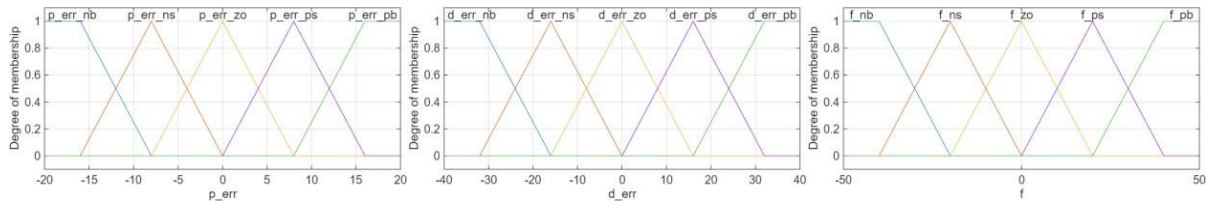


Figure 1: Mamdani_MFs

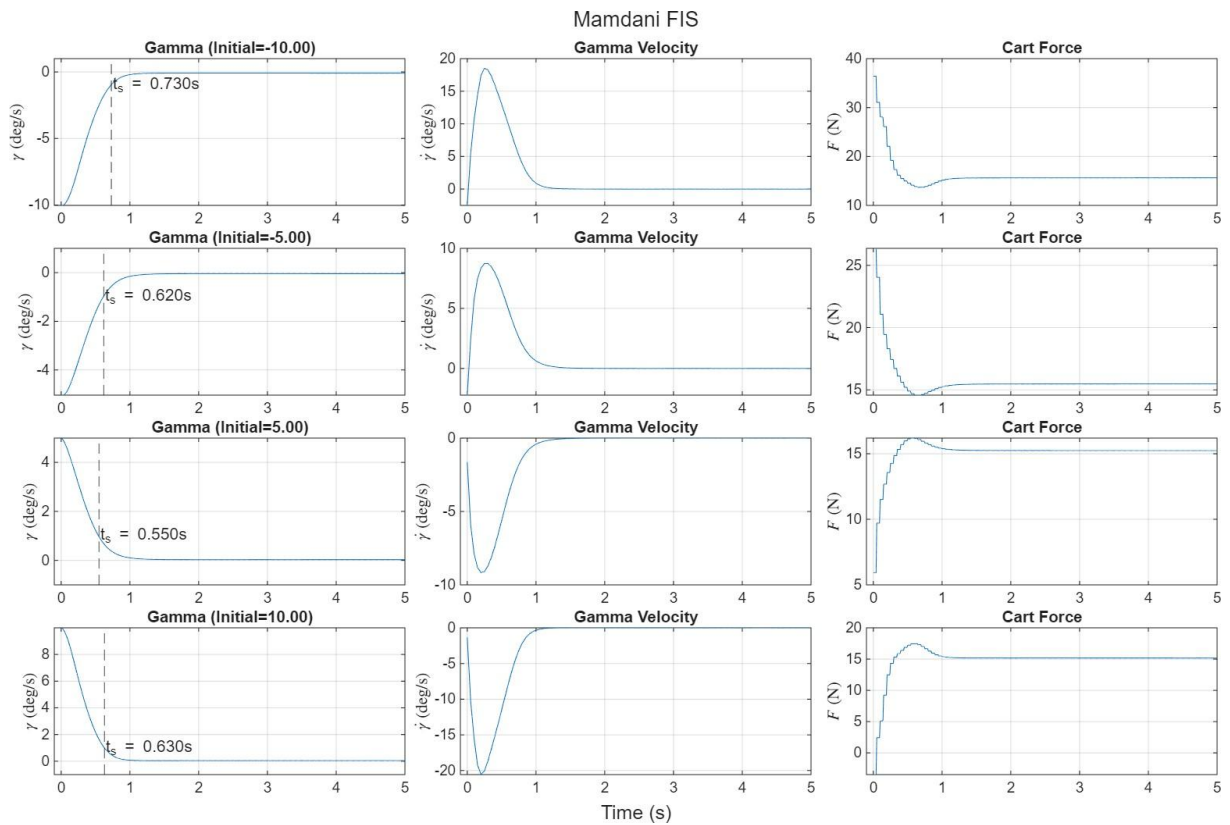


Figure 2: Mamdani_results

Sugeno System

Observations and Explanations

Serial No.	SIS Parts	Observations	Explanations
1.	Membership Functions (MFs)	Inputs: - p_err (position/angle error) range: [-20, 20] (deg) - d_err (derivative of angle) range: [-40, 40] (deg/s) - Each uses five trapezoidal membership functions (NB, NS, ZO, PS, PB). Output: - f (cart force) uses constant (singleton) MFs (f_nb, f_ns, f_zo, f_ps, f_pb) with crisp values (e.g. -39.17 N to +39.17 N).	Input overlap provides smooth rule activation: Overlapping triangles allow a single measurement to partially belong to multiple sets → smoother control and reduced chattering. Output constants (zero-order Sugeno) mean the control force is a weighted average of rule outputs → no centroid defuzzification. - Result: less sensitivity to output set geometry and lower overshoot compared to Mamdani, especially for large initial angles. Output singleton spacing controls control aggressiveness. - Wider spacing (e.g. ±40 N) → stronger corrective force, faster response but risk of actuator saturation.
2.	Inference Method	- AndMethod = 'min', OrMethod = 'max'. - Sugeno does not use aggregation of fuzzy output sets—the crisp output is computed directly as a weighted average of rule outputs.	Weighted averaging avoids centroid geometry issues. This yields a continuous control surface, leading to: - Lower overshoot under large disturbances. - Shorter settling time, because the force tracks error more proportionally.
3.	Bias Force (f_bias)	A slope compensation term $f_bias = g * \sin(\phi) * M_total$ is added to counteract the downhill gravitational component.	This feed-forward term removes the steady-state load, so the fuzzy controller only needs to correct dynamic deviations, improving steady-state accuracy and reducing continuous force output.
4.	Controller Execution Frequency	Controller updates every CTRL_STEP = 5 simulation steps.	A slower update acts as a low-pass filter on noise, but adds actuation delay.

		Simulation step is 0.01 s $\rightarrow$ control loop period = 0.05 s.	Delay slightly increases settling time compared to an ideal continuous controller but prevents high-frequency chattering.
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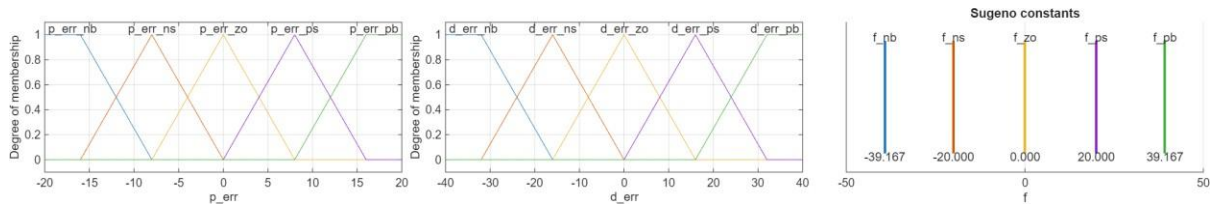


Figure 3: Sugeno_MFs

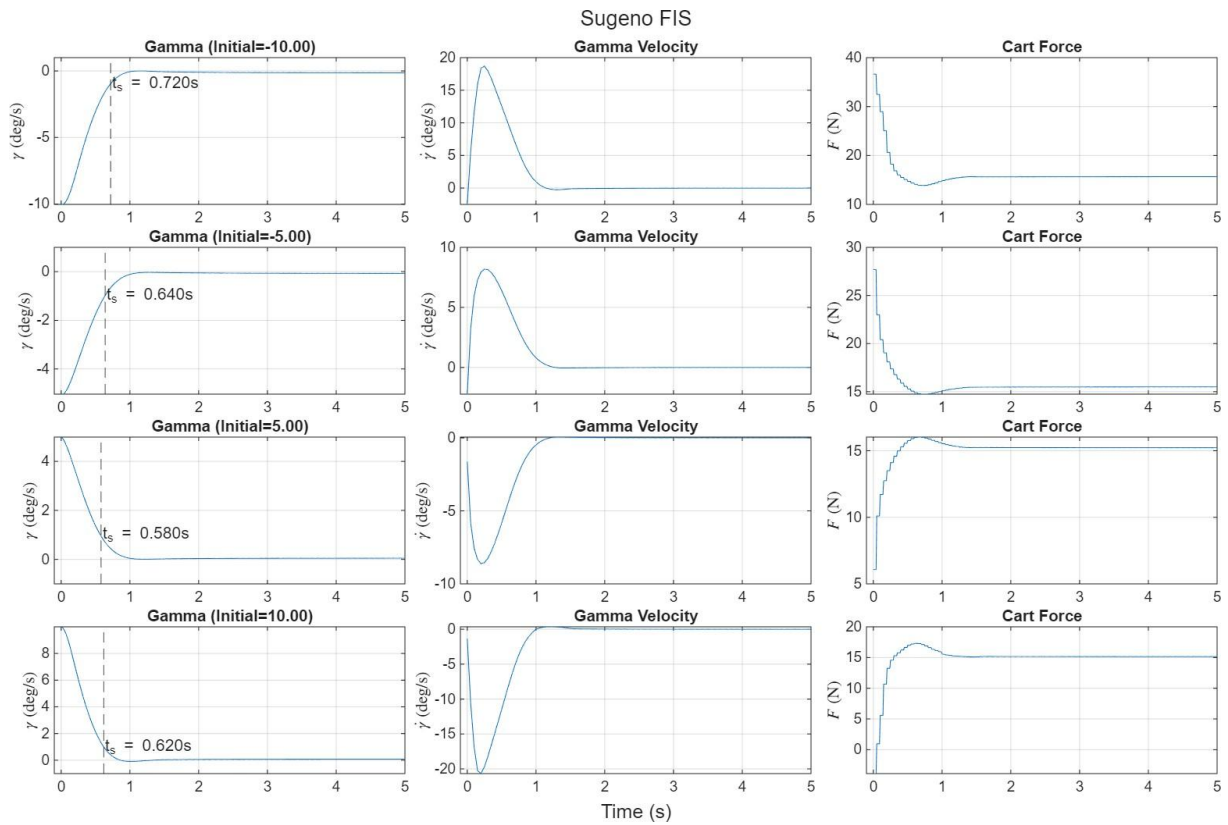


Figure 4: Sugeno_results

Common Observations

Serial No.	MIS and SIS Parts	Observations	Explanations
1.	Rule Base Structure	5×5 matrix mapping $p_err \times d_err$ to an output action. Diagonal rule dominance: Force direction opposes tilt and velocity to bring the pendulum upright. Gradual transitions across rules near the zero-error region.	- Diagonal symmetry provides stability by balancing corrective forces in all quadrants. - Smooth rule gradation ensures progressive force scaling, which is essential to avoid abrupt changes that cause overshoot.
2.	Input Scaling	Both codes scale the inputs (K_g for position, K_v for velocity) before feeding them into the fuzzy inference system.	Scaling gains determine relative sensitivity to angle vs. velocity: - Higher $K_g \rightarrow$ more aggressive tilt correction. - Lower $K_v \rightarrow$ less damping, faster but potentially more oscillatory response. Proper tuning of these gains balances settling time and overshoot.
3.	Bias/Feedforward Force	Both controllers include a bias (gravity compensation) term to counteract the downhill component of the slope	- This reduces the steady-state control effort, letting the fuzzy controller focus on dynamic deviations. - Improves steady-state accuracy and reduces continuous actuator stress.
4.	Control Execution & Saturation	Controllers run at a fixed control interval (e.g., every 0.05 s).	- Periodic updates introduce a small actuation delay, which acts as a low-pass filter on noise but slightly increases settling time. - Saturation prevents actuator damage but may lengthen recovery after large disturbances.

5.	Physical Constants & Saturation	- Force is clipped to ± 50 N (clip() function). - Random noise is added to the error inputs.	Force saturation limits excessive control action, improving safety but possibly increasing settling time if the required force exceeds 50 N. - Noise tests the robustness of the fuzzy controller. Overlapping MFs smooth the response and help maintain stability despite noise.
6.	Noise Handling	Small Gaussian noise is added to sensor readings to test robustness.	Overlapping MFs and rule redundancy smooth out the effect of noise, maintaining stable control without chattering.
7.	System Dynamics	- Both use Euler–Lagrange equations of a cart–pendulum on an inclined plane. - Settling time is measured when the angle stays within a tight band (e.g., $\pm 1^\circ$).	- Accurate plant modeling ensures that fuzzy rules are evaluated under realistic inertia, friction, and gravitational effects. - Both systems must manage the same physical challenges (delays, nonlinearities).

Rule Table

P_ERR	D_ERR					
		NB	NS	ZO	PS	PB
	NB	PB	PB	PS	PS	ZO
	NS	PB	PS	PS	ZO	NS
	ZO	PS	PS	ZO	NS	NS
	PS	PS	ZO	NS	NS	NB
	PB	ZO	NS	NS	NB	NB

Values Used in the Membership Function

P_ERR MFS

Parameters	Values
P_ERR_NB	[-20 -20 -16 -8]
P_ERR_NS	[-16 -8 -8 0]
P_ERR_ZO	[-8 0 0 8]
P_ERR_PS	[0 8 8 16]
P_ERR_PB	[8 16 20 20]

D_ERR MFS

Parameters	Values
D_ERR_NB	[-40 -40 -32 -16]
D_ERR_NS	[-32 -16 -16 0]
D_ERR_ZO	[-16 0 0 16]
D_ERR_PS	[0 16 16 32]
D_ERR_PB	[16 32 40 40]

Mamdani F MFS

Parameters	Values
F_NB	[-50 -50 -40 -20]
F_NS	[-40 -20 -20 0]
F_ZO	[-40 -20 -20 0]
F_PS	[0 20 20 40]
F_PB	[20 40 50 50]

Sugeno F MFS

Parameters	Values
F_NB	-39.1667
F_NS	-20.0000
F_ZO	0.0000
F_PS	20.0000
F_PB	39.1667

What happens when we change the membership functions:

1. Equal Distribution of MFs:

- Observation: Moderate settling time, small overshoot, smooth force that quickly decays to bias
- Why: Balanced overlap gives a moderate effective gain and damping; the controller is neither aggressive nor sluggish.

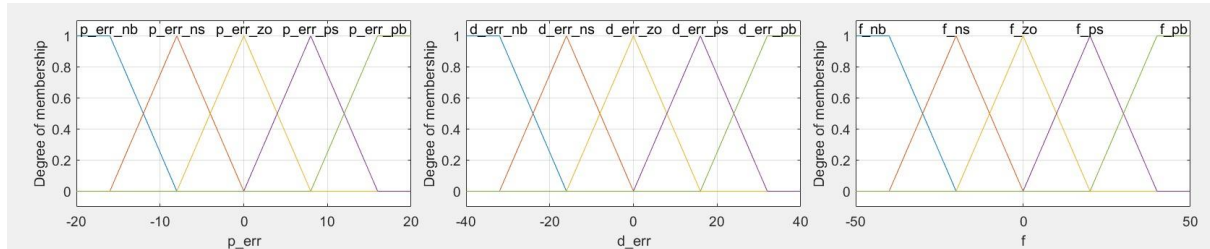


Figure 5: Membership Functions

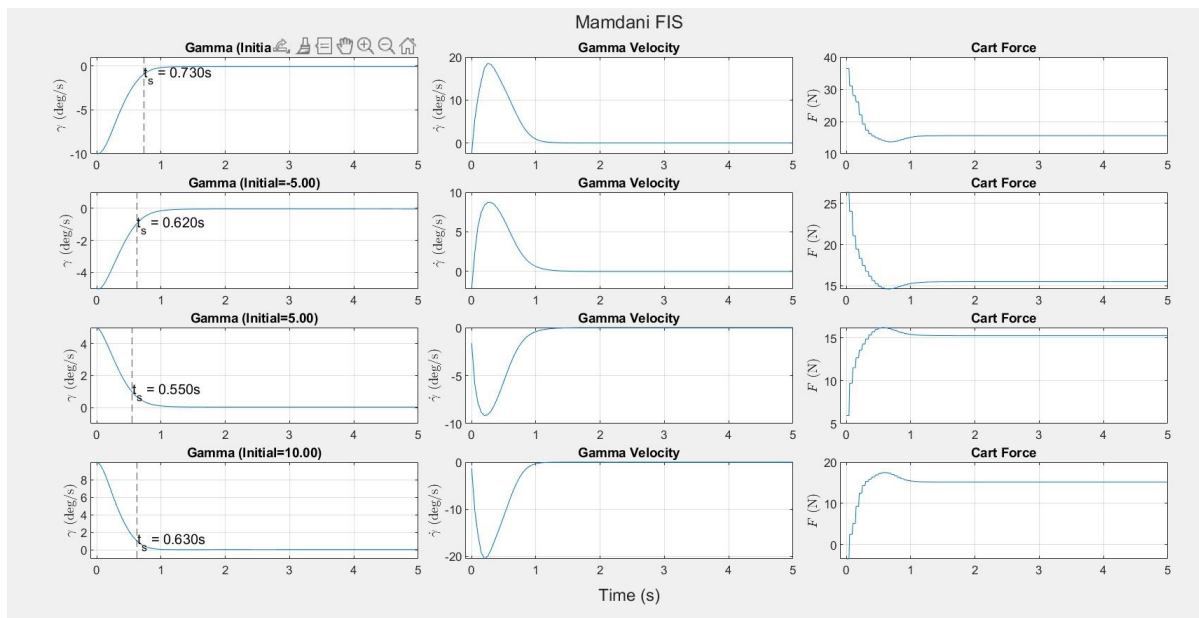


Figure 6: Results

2. Make only p_err ZO narrower (sharper around 0°):

- Observation: Settling time slightly reduced for small disturbances, Higher initial force spike
- Why: Shrinking the ZO zone makes the controller “jump out of neutral” earlier, so it reacts more aggressively to tiny errors (the controller switches to “positive/negative” correction sooner). That speeds up correction but can cause the pendulum to overshoot and wobble as damping hasn't increased.

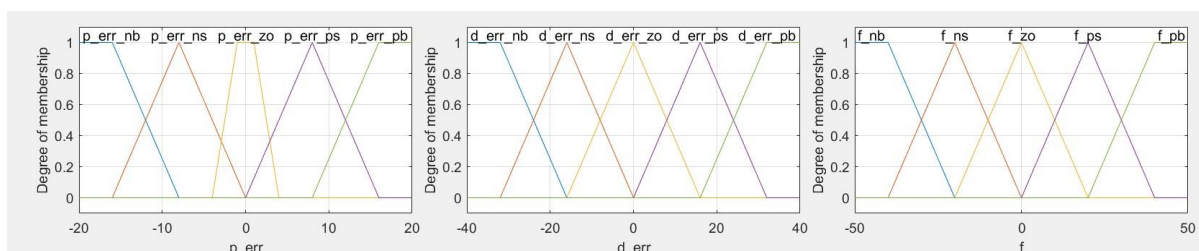


Figure 7: Membership Functions

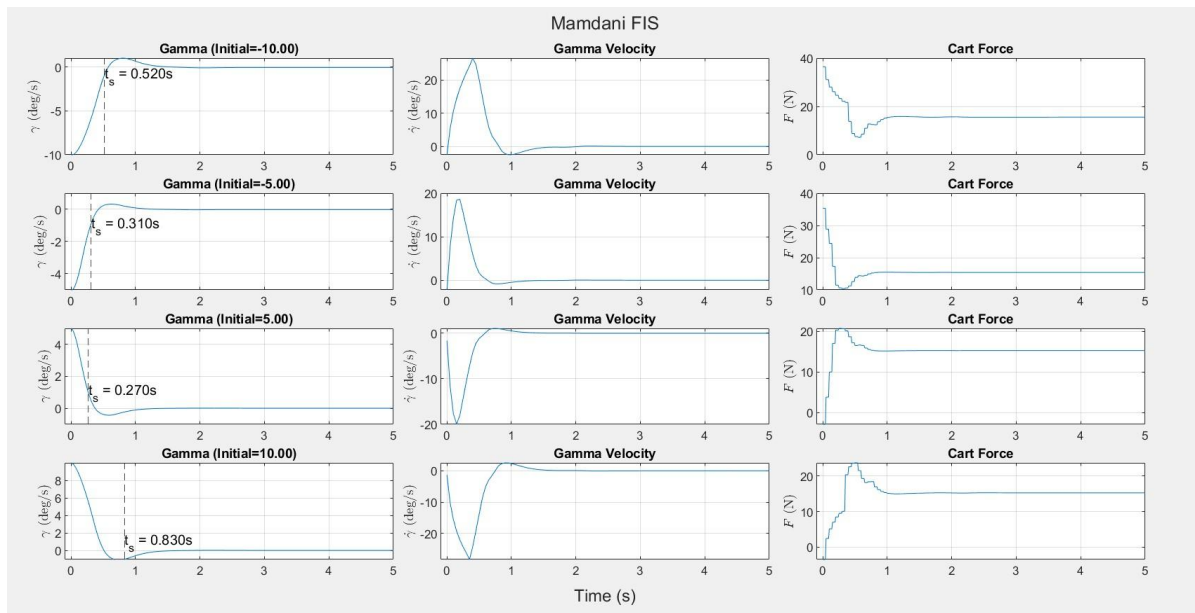


Figure 8: Results

3. Shift all p_err MFs right by +2°:

- Observation: Positive-angle disturbances corrected faster; negative-angle starts settle more slowly; final equilibrium slightly shifted to +2°.
- Why: Shifting MFs right makes the controller see the same angle as “more positive,” so it reacts earlier and harder for positive angles, but reacts later and weaker for negative angles. The internal zero is shifted, causing the physical equilibrium to move slightly.

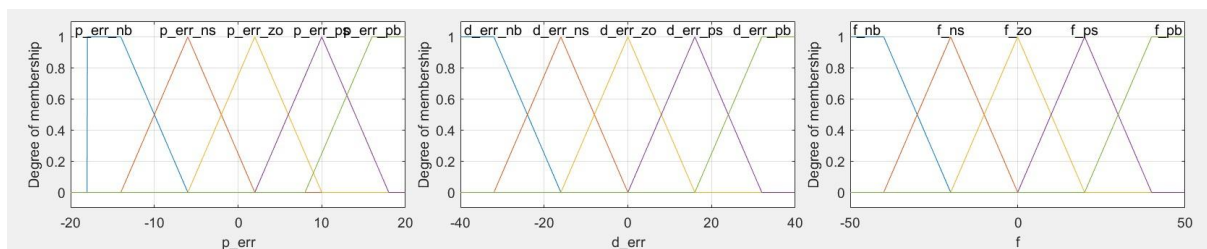


Figure 9: Membership Functions

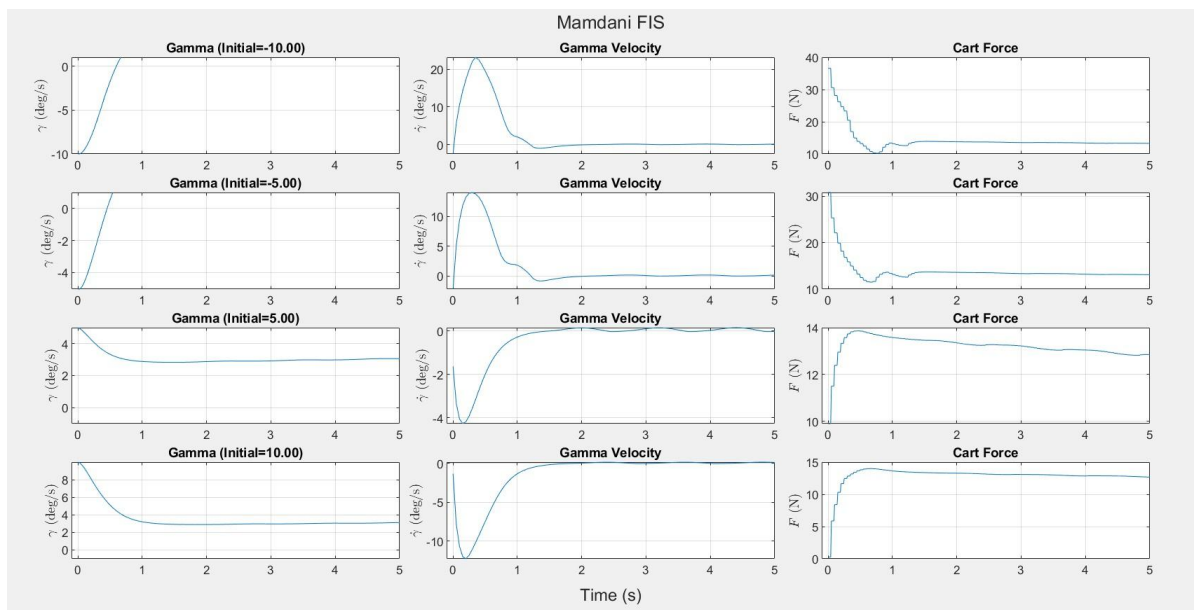


Figure 10: Results

4. Shift all d_err MFs left by $-4^\circ/s$:

- Observation: Positive-velocity swings slowed faster with less overshoot; negative-velocity swings take longer to settle, possibly overshooting more.
- Why: Shifting MFs left makes the controller interpret small positive velocities as “larger,” so it brakes earlier on that side. Small negative velocities appear “smaller,” so braking is weaker. This asymmetric response speeds up one side while slowing the other, creating uneven settling.

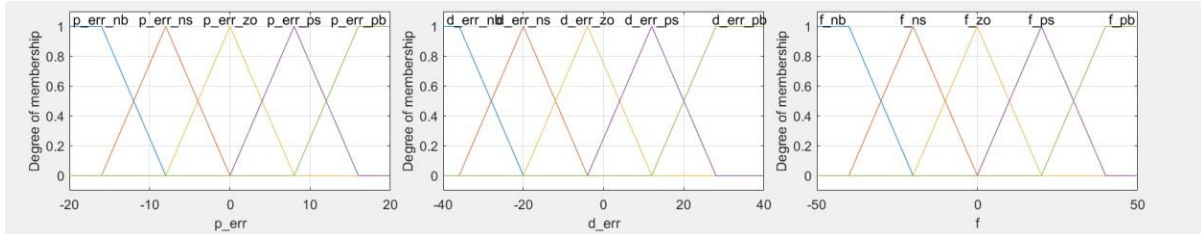


Figure 11: Membership Functions

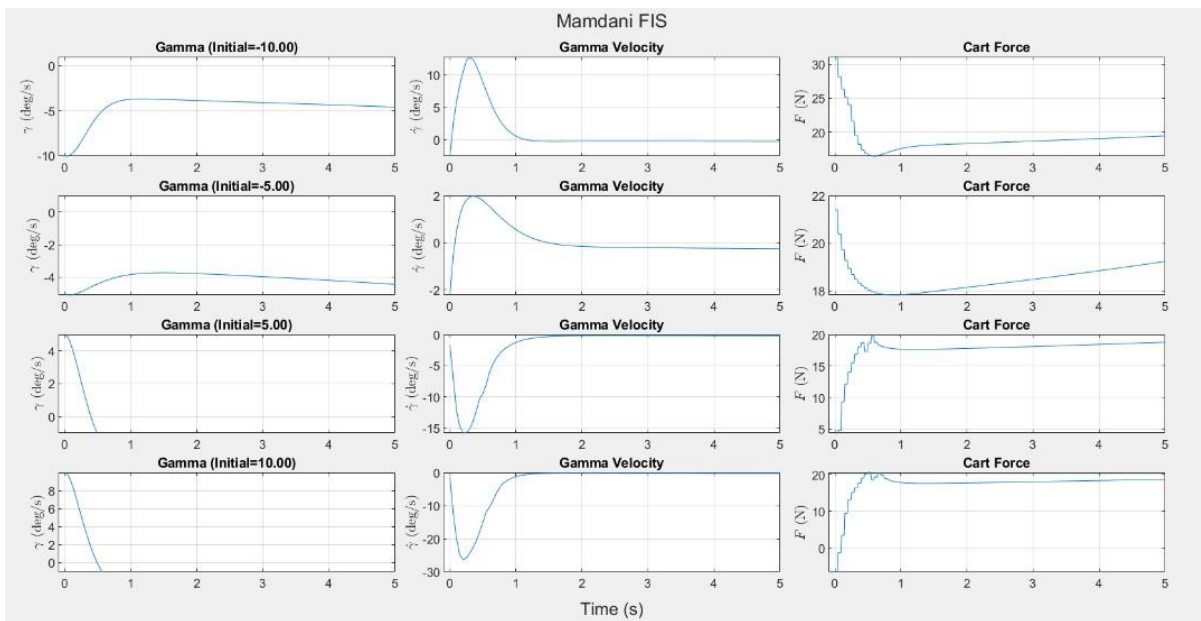


Figure 12: Results

5. Make p_err bands narrower (higher gain):

- Observation: Pendulum reacts faster, reaches upright quicker, but produces larger force spikes and may overshoot or wobble.
- Why: Narrower bands make the controller more sensitive—small angle changes trigger stronger corrections. This aggressive response speeds up settling but can hit force limits and induce oscillations.

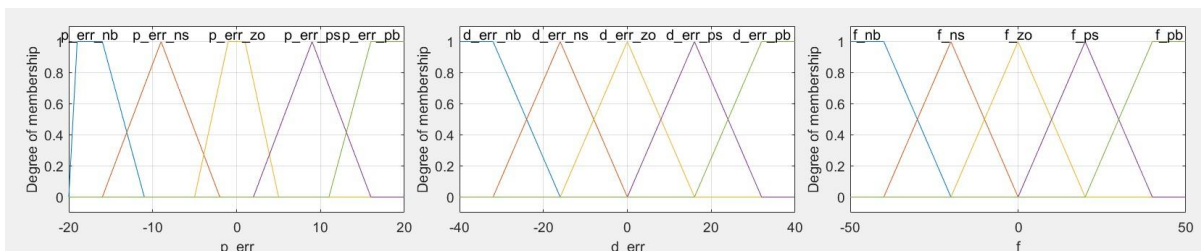


Figure 13: Membership Functions

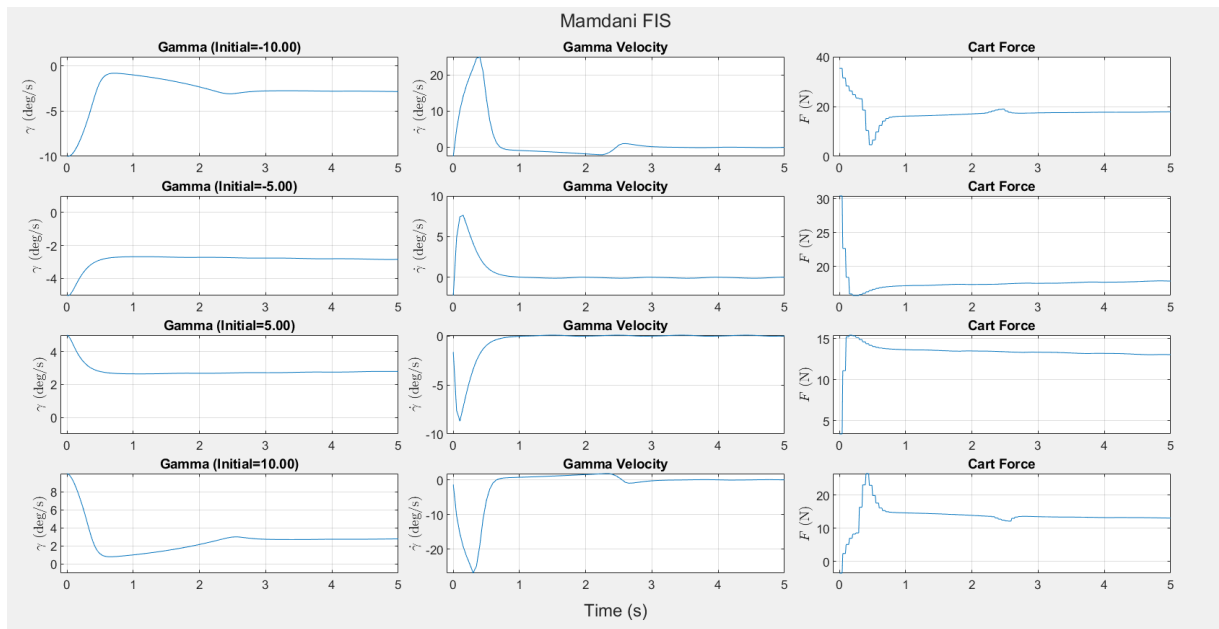


Figure 14: Results

6. Make d_err bands wider (softer derivative action $\rightarrow$ lower gain):

- Observation: Pendulum returns more gently; motion looks smoother; oscillations last longer before settling.
- Why: Wider bands reduce slope steepness, so moderate velocities trigger weaker braking. Weaker rule activation $\rightarrow$ smaller damping force, slowing pendulum correction and extending oscillation.

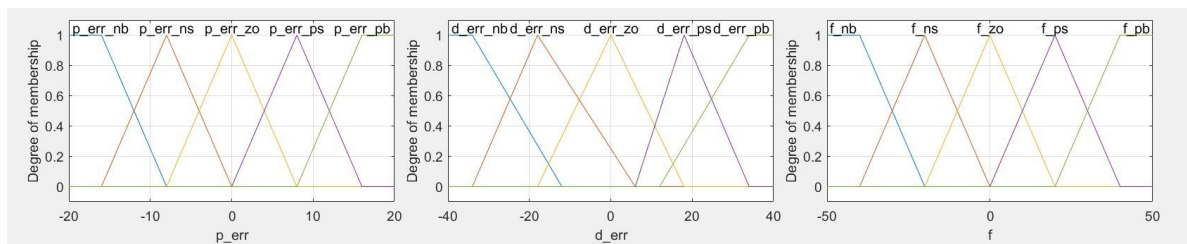


Figure 15: Membership Functions

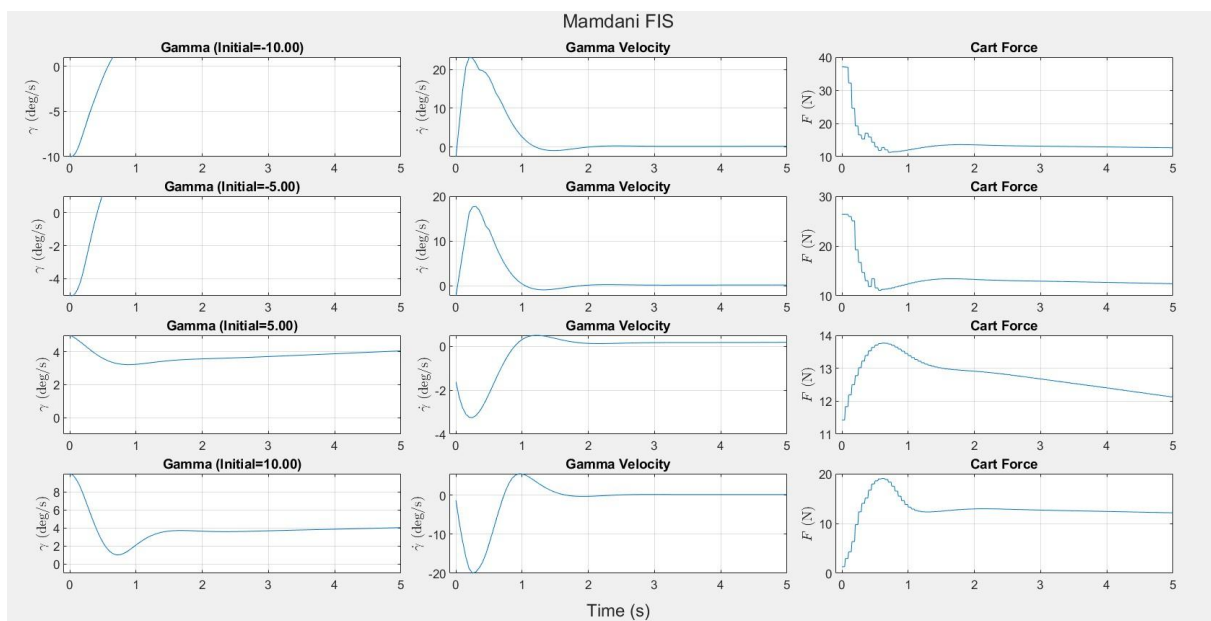


Figure 16: Results

7. Make output f narrower (higher effective control gain):

- Observation: Pendulum returns upright faster, stronger force spikes, more overshoot, can hit ± 50 N limit, sometimes unstable.
- Why: Narrower output MFs shift the centroid further from zero for the same activation, producing larger commanded forces. This amplifies the controller response, making it quicker but more aggressive and prone to oscillations.

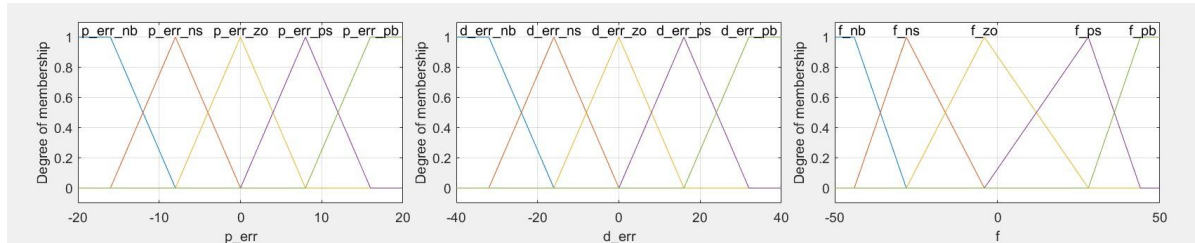


Figure 17: Membership Functions

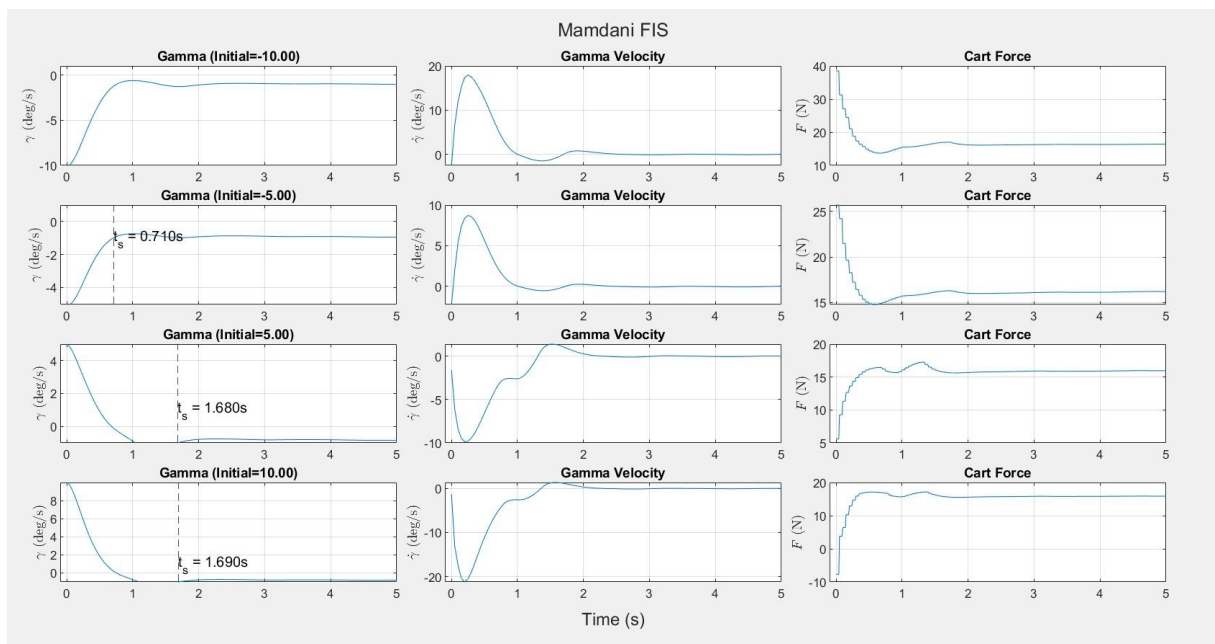


Figure 18: Results

8. Force about zero (symmetrical case):

- Observations: Cart force oscillates around zero; pendulum stabilizes symmetrically; force crosses zero repeatedly, showing reactive control without constant offset.
- Why: No static bias—force responds only to pendulum error; symmetric FIS rules and MFs generate positive/negative forces for corresponding errors; properly tuned gains (K_g , K_v) prevent saturation; frequent controller updates (CTRL_STEP) allow smooth, continuous adjustments.

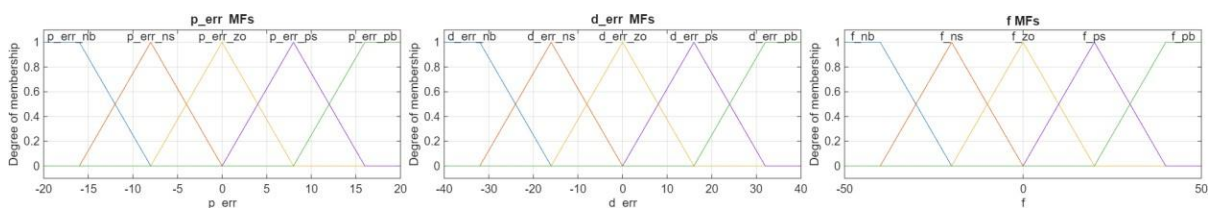


Figure 19: Membership Functions

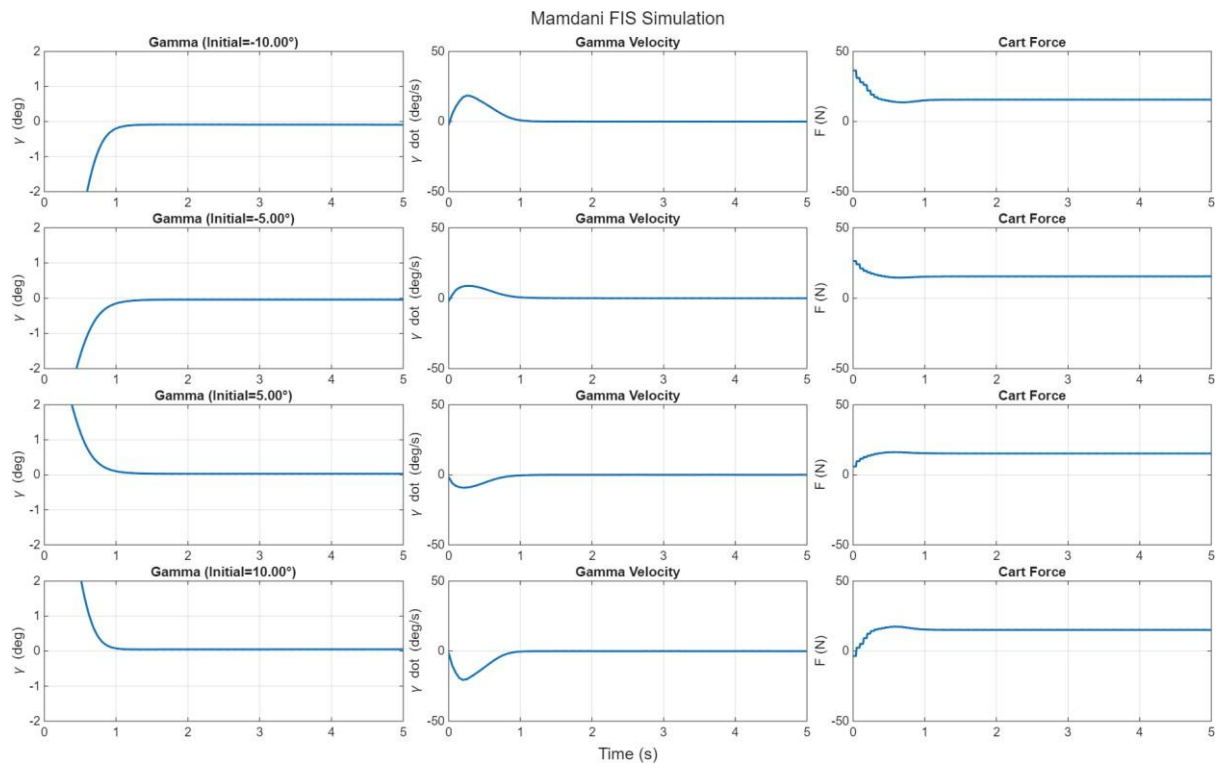


Figure 20: Results

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